

The Motivational Role of the Ventral Striatum and Amygdala in Learning From Gains and Losses

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The ventral striatum (VS) and amygdala are two structures often implicated as essential structures for learning. The literature addressing the contribution of these areas to learning, however, is not entirely consistent. We propose that these inconsistencies are due to learning environments and the effect they have on motivation. To differentiate aspects of learning from environmental factors that affect motivation, we ran a series of experiments with varying task factors. We compared monkeys (*Macaca mulatta*) with VS lesions, amygdala lesions, and unoperated controls on reinforcement learning (RL) tasks that involve learning from both gains and losses as well as from deterministic and stochastic schedules of reinforcement. We found that for all three groups, performance varied by experiment. All three groups modulated their behavior in the same directions, to varying degrees, across the three experiments. This behavioral modulation is why we find deficits in some experiments, but not others. The amount of effort animals exhibited differed depending on the learning environment. Our results suggest that the VS is important for the amount of effort animals will give in rich deterministic and relatively leaner stochastic learning environments. We also showed that monkeys with amygdala lesions can learn stimulus-based RL in stochastic environments and environments with loss and conditioned reinforcers. These results show that learning environments shape motivation and that the VS is essential for distinct aspects of motivated behavior.

Keywords: ventral striatum, amygdala, reinforcement learning, motivation, schedules of reinforcement

Reinforcement learning (RL), a computational approach for learning how to map situations to actions, attempts to maximize reward and minimize punishment. One of the most successful biological RL theories, temporal difference RL, suggests that midbrain dopamine codes the temporal difference reward prediction error (RPE; Sutton & Barto, 1998), which is the difference between expected reward and actual reward (Houk et al., 1995; Schultz et al., 1997). This theory has led to the idea that the ventral striatum (VS) and amygdala, which receive a prominent dopamine projection, are essential structures for learning the value of consequences that follow actions (Averbeck & Costa, 2017; Averbeck & O'Doherty, 2022; Costa et al., 2016).

Although this model explains a wealth of data, the literature concerning the role of these structures has inconsistencies. The type of learning the VS is thought to underlie most closely resembles instrumental conditioning, but when this has been tested the results have been inconsistent. For example, VS lesions disrupt instrumental conditioning in some tasks (Costa et al., 2016; Rothenhoefer et al., 2017), but not others (B. Balleine & Killcross, 1994; Vicario-

Feliciano et al., 2017). In addition, even when these lesion studies have found deficits, the magnitude of deficits varies (Costa et al., 2016; Rothenhoefer et al., 2017; Taswell et al., 2018). Similarly, some studies have shown that amygdala lesions produce deficits in instrumental conditioning (Costa et al., 2016; Rudebeck et al., 2017; Rudebeck & Murray, 2008), while others have found no deficits (B. W. Balleine et al., 2003; Izquierdo & Murray, 2007; Málková et al., 1997).

Furthermore, the RPE theory of dopamine also predicts that increasing or decreasing the phasic dopamine signal should increase or decrease the RPE signal and correspondingly increase or decrease learning rates. Learning rates affect the rate at which asymptotic behavioral performance is approached (B. B. Averbeck & Costa, 2017). Contrary to the RPE theory of dopamine, behavioral pharmacology studies often find effects on choice consistency or asymptotic performance and not effects on learning rate (Costa et al., 2015; Eisenegger et al., 2014; Lee et al., 2015; Pessiglione et al., 2006). Maintaining consistent choices requires using values to make

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decisions (motivated behavior or performance) rather than updating values (learning). Several reviews of the literature concerning dopamine and these two structures have found that dopamine systems are not needed to learn the value of reinforcers, but instead important for wanting reinforcers (Berridge & Robinson, 1998; Cardinal et al., 2002). Thus, dopamine, the VS, and the amygdala may be important for motivation, and only important for learning behavior–outcome contingencies via motivation.

There is an extensive literature that suggests that dopamine influences motivated behavior (Hamid et al., 2016; Howe et al., 2013; Phillips et al., 2003; Satoh et al., 2003; Wassum et al., 2011). When this view of dopamine has been tested, work has found that dopamine is important for the amount of effort animals are willing to expend for a particular task (Cocker et al., 2012; Hamid et al., 2016; Niv et al., 2007; Shenhav et al., 2017; Westbrook & Braver, 2016; Westbrook et al., 2020). There is also work exploring the idea that tonic dopamine conveys motivational value (Niv et al., 2007) and phasic dopamine conveys RPEs for learning (Berke, 2018). This view of dopamine unifies and explains an otherwise conflicting literature.

We hypothesize that important details in the behaviors studied in the learning experiments, which affect levels of motivation, have led to some of the conflicts in the literature. For example, many of the studies that find a role for dopamine, the VS, or the amygdala in learning employ stochastic concurrent appetitive reward structures (Costa et al., 2016; Eisenegger et al., 2014; Pessiglione et al., 2006; Rothenhoefer et al., 2017; Taswell et al., 2018, 2021). When tasks have employed deterministic reward structures, these effects are diminished or disappear completely (Costa et al., 2016; Málková et al., 1997; Taswell et al., 2018; Vicario-Feliciano et al., 2017). There is a large behavioral literature showing that reward rate, reward schedule, and reward magnitude affect performance (Skinner, 1953; Skinner & Ferster, 2015).

RL theories have been shown to be reward maximizing under some conditions (Sutton & Barto, 1998). However, biological agents often deviate from optimality, although, as has been pointed out, this is often overlooked in the literature (Sutton & Barto, 1998). Separating learning rates from motivational factors can be difficult. To understand and control for environmental effects in learning tasks, we must run a series of tasks, where each task varies by one parameter. This is how we can separate the motivational effects of the environment (task) from learning ability, which can allow for clearer statements about the role of the VS and amygdala in learning.

In the present study, we conducted additional analyses on previously published data (Taswell et al., 2018) and compared this to new behavioral data. We ran a series of tasks testing monkeys with lesions of the VS against unoperated controls and found that monkeys with lesions of the VS had performance deficits in two out of four RL tasks. The present study used the same unoperated controls (Taswell et al., 2018), but compared them to monkeys with lesions of the amygdala. We found that monkeys with amygdala lesions had no performance deficits in any of the four tasks. In these tasks, rhesus monkeys made choices among options and received tokens for their choices. The tokens were represented by circles on the bottom of the screen and the animals periodically received juice in exchange for accumulated tokens. With this task, we aim to differentiate aspects of learning stimulus–outcome associations from task factors that affect motivation by evaluating the behavioral performance of three rhesus macaques with VS lesions, four rhesus

macaques with amygdala lesions, and four unoperated controls across three of the four token task variants (Taswell et al., 2018).

Method

Subjects

The subjects included nine male and two female rhesus macaques with weights ranging from 6 to 11 kg. Three male monkeys and one female monkey received bilateral excitotoxic lesions of the amygdala and were the same monkeys used in a previous study (Taswell et al., 2021). Three male rhesus macaques received bilateral excitotoxic lesions of the VS and were the same monkeys used in a previous study (Costa et al., 2016; Rothenhoefer et al., 2017; Taswell et al., 2018). The remaining four monkeys, three males and one female, served as unoperated control monkeys and were the same monkeys used in a previous study (Taswell et al., 2018). One of these control monkeys were used in Taswell et al. (2021). Two control subjects were new naïve monkeys whose first task was these token experiments. The final control monkey had been trained on a sequence learning task before starting the token experiments. One male from both the control and amygdala lesion animal group was not able to complete the final task (TkL), and therefore, the fourth task (TkL) only has three controls and three amygdala lesion monkeys. For the duration of the study, monkeys were placed on water control. On testing days, monkeys earned their fluid from their performance on the task. Experimental procedures for all aspects of the study were performed in accordance with the *Guide for the Care and Use of Laboratory Animals* and were approved by the National Institute of Mental Health Animal Care and Use Committee.

Surgery

Four monkeys received two separate stereotaxic surgeries, one for each hemisphere, which targeted the amygdala using the excitotoxin ibotenic acid (for details, see Taswell et al., 2021). Three other monkeys received two separate stereotaxic surgeries, one for each hemisphere, which targeted the VS using quinolinic acid (for details, see Costa et al., 2016; Taswell et al., 2018) which lesioned both the VS core and shell. Injection sites were determined based on structural magnetic resonance scans obtained from each monkey prior to surgery. After both lesion surgeries had been completed, each monkey received a cranial implant of a titanium head post to facilitate head restraint. Unoperated controls received the same cranial implant. Behavioral testing for all monkeys began after they had recovered from the implant surgery.

Lesion Assessment

Lesion volume estimates were taken by first transforming each subject's T2-weighted scan acquired 1 week postoperatively to the standard National Institute of Mental Health macaque template (NMT; Seidlitz et al., 2018) using Analysis of Functional Neuro-Images's 3dAllineate function (Cox, 1996). We then applied thresholding to identify the area of hyperintensity on the transformed T2-weighted object to isolate a binary mask that corresponded to the area of damage. The masks were visually inspected and manually edited to ensure that they fully captured the areas of hyperintensity on the T2-weighted object. A lesion overlap map was created by summing the binary masks for each hemisphere and displaying the output on the NMT. For VS monkeys, see Costa et al. (2016;

(Taswell et al., 2018), and for amygdala monkeys, see Taswell et al. (Taswell et al., 2021).

Task

We conducted a post hoc analysis on data from the token experiments reported in Taswell et al. (2018). We use data from only three of the four token experiments, the first, the third, and the fourth. Whereas the first, third, and fourth token tasks each have four cues and blocks that consist of 108 trials, which means monkeys saw each condition 18 times, the second token task had five cues and a block length of 120 trials, which means monkeys saw each condition 12 times. Because the differences in block length and number of cues makes it difficult to compare learning across tasks, we excluded the second task from this analysis.

TkD—Token Task 1 (Deterministic Learning)

In the first task (TkD), novel blocks consisted of 108 trials and familiar blocks of 36 trials. Novel blocks consisted of four images the animals had never seen before. Associated with each image was a value (+2, +1, -1, -2), such that if that image was chosen, the animal gained or lost the corresponding number of tokens. On each trial, monkeys had to acquire and hold central fixation for 500 ms. After monkeys held central fixation, two of the images would appear to the left and the right of the fixation point. The animal chose one by making a saccade to the image and holding for 500 ms. The number of tokens associated with the image was then added or subtracted from their total count, represented by circles at the bottom of the screen. The animals could not have less than zero tokens, however. Therefore, if they had one token and they chose a -2 image, they were reduced to zero tokens. Every four to seven trials their tokens were cashed out. At cash-out, the animals were given one drop of juice for each token. When each drop of juice was delivered, one token was removed from the screen. There were six individual conditions in this task, defined by the possible pairs of images. The conditions within a block of 108 trials were presented pseudorandomly. The animals saw each condition twice, once on the left and once on the right, every 12 trials before seeing any condition a third time. At the end of each 108-trial block, we introduced four new images and the animals began the learning from scratch.

TkS—Token Task 3 (Stochastic Learning)

In the third task (TkS), we examined performance when feedback was stochastic. In this task, at the beginning of each block, we introduced four new images with associated reward magnitudes of +2, +1, -1, and -2. The design was otherwise the same as the first token task, with the exception that rewards were stochastic. Specifically, in 75% of the trials, the number of tokens was adjusted by the magnitude associated with the chosen option, but in 25% of the trials, there was no change in the number of tokens.

TkL—Token Task 4 (Deterministic Learning)

In the final task (TkL), we again used deterministic feedback to examine performance. This version of the task was like task 1 (TkD) with two differences. First, we changed the value of the -2 cue to -4. So, the cues for this task were +2, +1, -1, -4. Second, we gave

the animals an endowment of four tokens after every cash-out to maintain motivation and to increase the number of trials on which they would experience the actual four-token loss.

Images and Eye Tracking

Images provided as choice options were normalized for luminance and spatial frequency using the SHINE toolbox for MATLAB (Willenbockel & Sadr, 2010). All images were converted to grayscale and subjected to a two-dimensional fast Fourier transform to control spatial frequency. To obtain a goal amplitude spectrum, the amplitude at each spatial frequency was summed across the two image dimensions and then averaged across images. Next, all images were normalized to have this amplitude spectrum. Using luminance histogram matching, we normalized the luminance histogram of each color channel in each image, so it matched the mean luminance histogram of the corresponding color channel, averaged across all images. Spatial frequency normalization always preceded the luminance histogram matching. Each day before the monkeys began the task, we manually screened each image to verify its integrity. Any image that was unrecognizable after processing was replaced with an image that remained recognizable. Eye movements were monitored, and the image presentation was controlled by PC computers running the Monkeylogic (Version 1.1) toolbox for MATLAB (Asaad & Eskandar, 2008) and Arrington ViewPoint eye-tracking system (Arrington Research).

RL Models

We fit a Rescorla–Wagner or stateless RL value update equation given by the following:

$$v_i(t+1) = v_i(t) + \alpha_j(R - v_i(t)), \quad (1)$$

These values were then passed through a soft-max function to give choice probabilities for the pair presented in each trial:

$$d_j(t) = (1 + e^{\beta_k(v_i(t) - v_j(t) + h_i(t) - h_j(t))})^{-1}, d_i(k) = 1 - d_j(k). \quad (2)$$

The variable v_i is the value estimate for option i , R is the change in the number of tokens that followed the choice in trial t , and α_i is the cue-dependent learning rate parameter, for cue i . Our cue model consisted of one learning rate parameter for each cue. In addition, we also used one choice consistency parameter, β_k , that was fit across all six conditions. The variable $h_i(t)$ implemented a choice autocorrelation function (Gershman et al., 2009), which increased the value of a cue that had occurred in the same location, recently. The autocorrelation function was defined as:

$$h_i(t) = \kappa e^{-\lambda(t - t_{li})}, \quad (3)$$

where the variables, κ and λ , were free parameters scaling the size of the effect and the decay rate, respectively. The variable, t_{li} , indicates the last trial on which a given cue, i , was chosen in a given location. There were eight separate values for t_{li} as it tracked the four cues across locations.

We then maximized the likelihood of the animal's choices, D , given the parameters, using as a cost function:

$$f(D|\alpha_j, \beta_k) = \prod_t [d_1(k)c_1(k) + d_2(k)c_2(k)]. \quad (4)$$

where $c_1(k)$ was an indicator variable that took on a value of 1 if Option 1 was chosen and 0 otherwise and $c_2(k)$ took on a value of 1 if Option 2 was chosen and 0 otherwise.

Statistics

To quantify differences between choice behavior in each group, we performed an arcsine transformation on the choice accuracy values, as this transformation normalizes the data (Zar, 1999). We then carried out an N -way analysis of variance. All single condition choice analyses are presented in their uncorrected form. Monkey and session were included as random effects with session nested under monkey. All other factors were fixed effects. Within-group analysis was done the same way, the group factor was dropped.

Results

We adapted a token reward system to two-armed bandit RL tasks (Taswell et al., 2018). The tasks involved a choice between two images (Figure 1A). The selection of an image led to an increase or decrease of tokens. Circles on the bottom of the screen represented tokens. Every four to seven trials, with an interval chosen randomly, accumulated tokens were exchanged for juice reward, and one drop of juice was delivered for each token. At the start of each 108-trial block, animals were introduced to four novel images. In each block, there were six conditions, defined by all the possible cue combination. Although different cue combinations make up the conditions, cues and their corresponding value were the same across conditions throughout a block. The use of tokens, which are secondary reinforcers, allowed us to study the effects of gains and losses on choices using the same unit of value.

We ran four tasks in a fixed sequence. The first was a deterministic tokens task (TkD) with a fixed, deterministic gain or loss of tokens following each choice (+2, +1, -1, or -2). In this article, we omit any consideration of the second task because it contained additional stimulus–outcome associations and had a shorter block of trials. The third was a stochastic version of the tokens task (TkS) with a 75% chance of receiving the same set amount of token gains or losses and a 25% chance of receiving no tokens, which changes the reward schedule and rate. The fourth tokens task (TkL) is a deterministic task with a large loss of four tokens (+2, +1, -1, or -4) and an initial endowment following each cash-out, which changes reward magnitude and rate (Figure 1A).

Average Performance Across Tasks

We started by comparing the data for the deterministic, stochastic, and large loss token experiments (Figure 1B–D). When we compared the average percent of correct choices monkeys made, where correct was defined as choosing the more valuable option (Figure 2D), we found that performance differed across experiments, experiment; $F(2, 14) = 33.1, p < .001$, and that group performance differed by experiment, Experiment $\times$ Group; $F(4, 14) = 3.6, p = .032$. To see which groups were driving this effect, we completed pairwise analysis of the groups across the three experiments. When we compared controls to VS monkeys, we found a marginal effect for the groups to differ across experiments, Experiment $\times$ Group; $F(2, 9) = 6.6, p = .017$, uncorrected. When we compared control and amygdala

monkeys, we found no differences between the groups, Experiment $\times$ Group; $F(2, 10) = 0.2, p = .827$, and when we compared amygdala and VS monkeys, we found a trend for groups to differ across experiments, Experiment $\times$ Group; $F(2, 9) = 4.7, p = .038$, but this effect did not survive correction.

To see the extent to which each group performed differently across the experiments, we completed separate within-group analysis for each group across experiments (Figure 2A–C). For the control and amygdala monkeys, we found that performance varied across experiments, controls: $F(2, 5) = 39.2, p = .001$; amygdala: $F(2, 5) = 21.2, p = .005$. For the VS monkeys, we found that performance did not differ by experiment, experiment; $F(2, 4) = 5.4, p = .073$. Thus, the control and amygdala groups performed differently across experiments, but the VS monkeys did not.

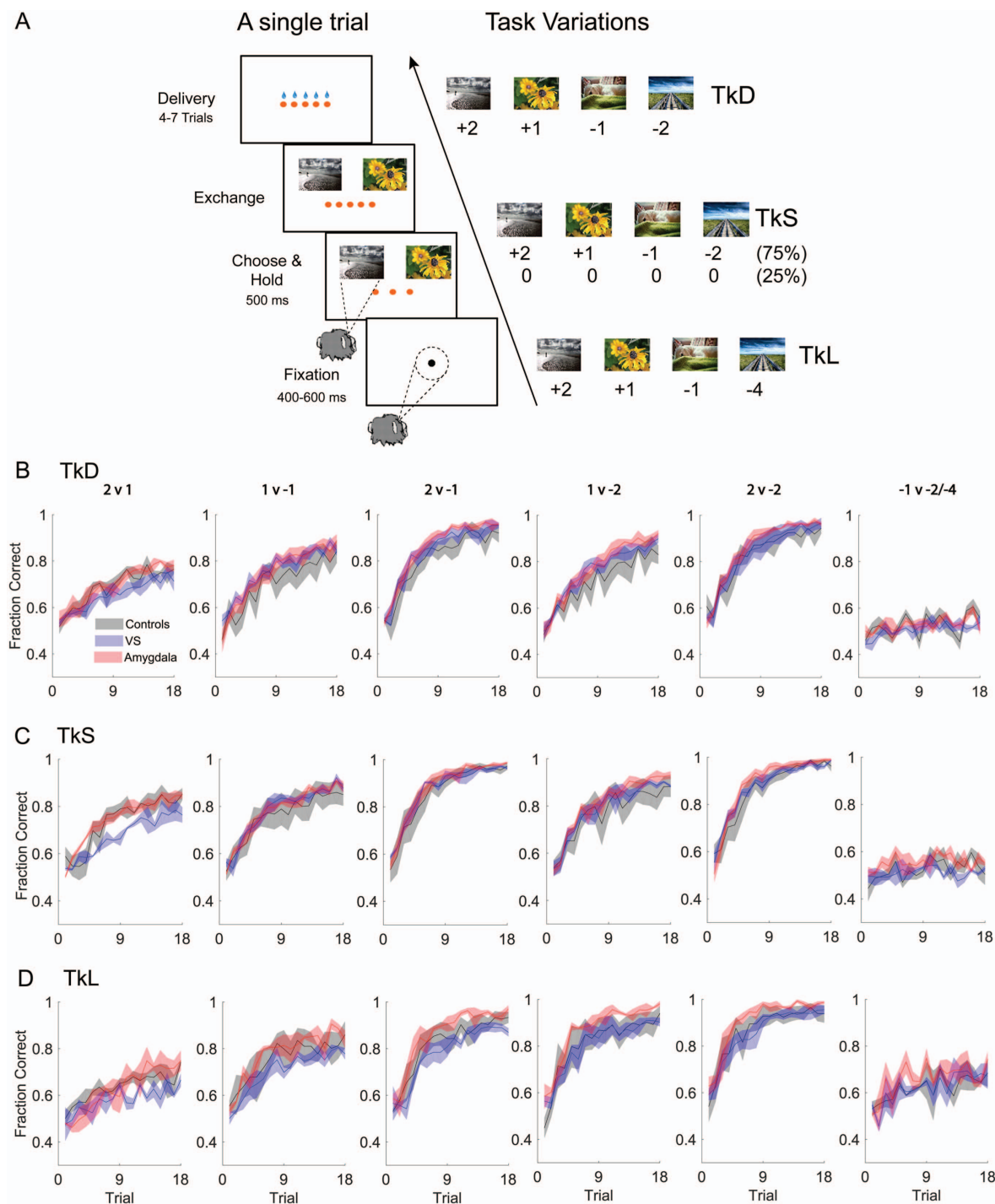
Next, we examined the pairwise differences between experiments. When we examined the performance for all three groups across TkD and TkS, we found that monkeys performed better in the TkS, experiment; $F(1, 8) = 33, p < .001$. There were no significant group effects in this comparison. However, when we performed within-group analysis across TkD and TkS, we found that control and amygdala monkeys performed better in TkS, Controls: $F(1, 3) = 24.5, p = .016$; Amygdala: $F(1, 3) = 23.2, p = .017$. This was not the case for VS monkeys, Experiment; $F(1, 2) = 10.2, p = .085$. Thus, the within-group analyses revealed that control and amygdala monkeys more often selected the better option in TkS when compared to TkD, but this was not the case for VS monkeys.

When we examined the performance for all three groups between TkS and TkL, we found no experiment or group effects. When we examined the performance for all three groups between TkD and TkL, we found that monkeys performed better in TkL, experiment; $F(1, 6) = 53.1, p < .001$. We also found that the groups differed by experiment, Experiment $\times$ Group; $F(2, 6) = 11.7, p = .008$. When we compared control to VS monkeys, we found that the groups differed between experiments, Experiment $\times$ Group; $F(1, 4) = 27.9, p = .006$. When we compared control and amygdala monkeys, we found no differences between the groups, Experiment $\times$ Group; $F(1, 4) = 0.2, p = .654$. In the final group comparison of amygdala and VS monkeys, we found that group performance differed between experiments, Experiment $\times$ Group; $F(1, 4) = 16.4, p = .015$.

We next carried out within-group analysis between TkD and TkL. This analysis revealed that control and amygdala monkeys performed better in TkL, controls: $F(1, 2) = 38.9, p = .024$; amygdala: $F(1, 2) = 22.4, p = .042$. This was not the case for VS monkeys, experiment; $F(1, 2) = 0.2, p = .718$. Thus, this within-group analysis revealed that VS monkeys treated TkD and TkL the same, while control and amygdala monkeys more often selected the better option in TkL when compared to TkD.

RL Model

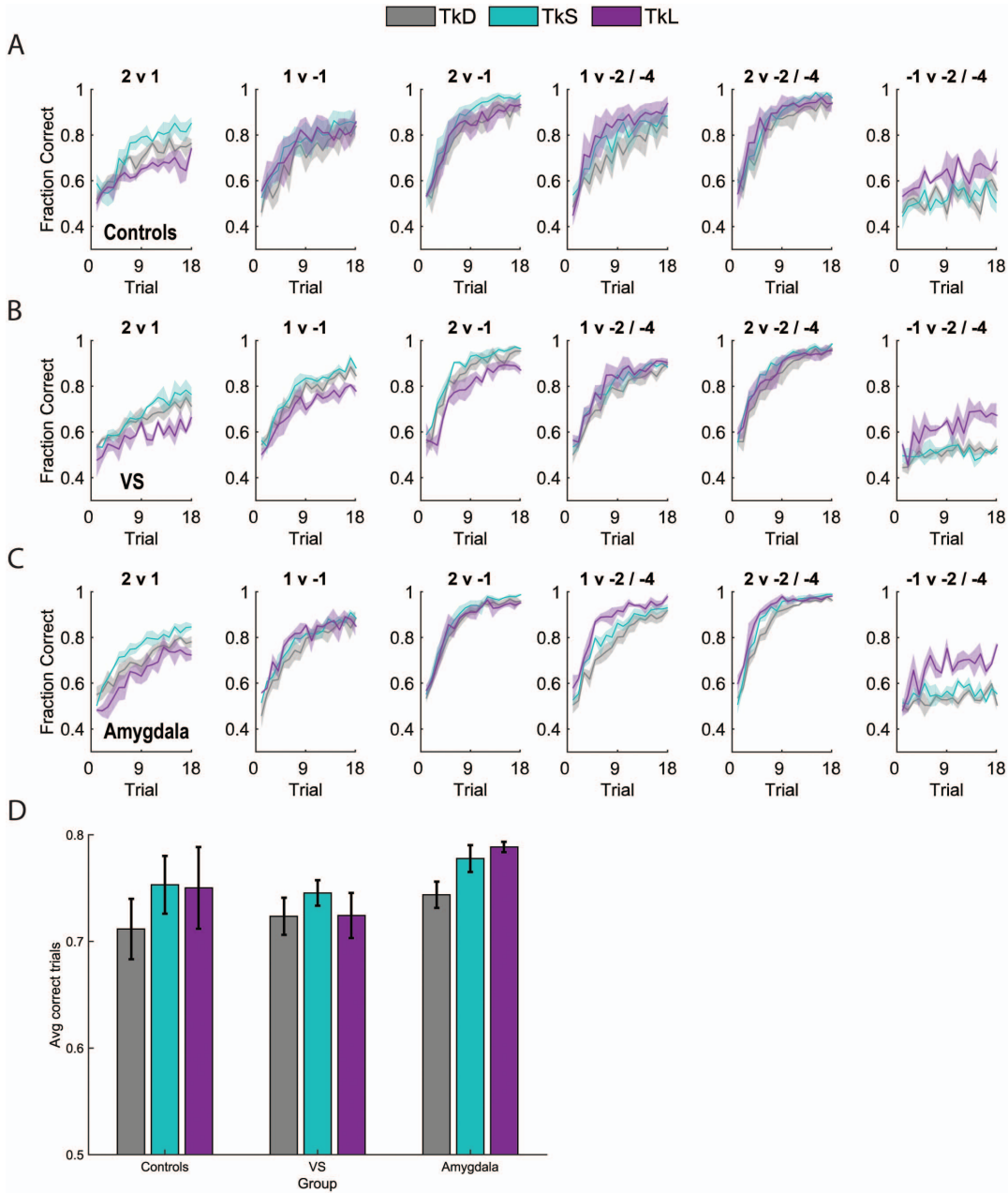
We previously found that a model consisting of one learning rate for each cue best predicted behavior (Taswell et al., 2018). In addition to the learning rate parameters, this model had one inverse temperature parameter across all conditions, one choice autocorrelation parameter, and one decay parameter (Taswell et al., 2018). The inverse temperature parameter is a measure of choice consistency. Larger values indicate that monkeys picked the higher value cue more often. The autocorrelation parameter increased the value of

Figure 1*Task Figure and Across-Group Performance Across the Three Tasks*

Note. (A) Schematic of the two-armed bandit RL tokens tasks. (B–D) Across-group behavioral performance across the three tasks. (A) The tasks involved a choice between two images and the selection of an image led to an increase or decrease of tokens. The circles on the bottom of the screen represented tokens. Every four to seven trials, with an interval chosen randomly, accumulated tokens were exchanged for juice reward. One drop of juice was delivered for each token. Three variations of the task were used. First, a deterministic tokens task (TkD) with a fixed, deterministic gain or loss of tokens (+2, +1, -1, or -2). A stochastic version of the tokens task (TkS) was used, which had a 75% chance of receiving the same set amount of token gains or losses and a 25% chance of receiving no tokens. Last, we ran a deterministic task with a large loss (TkL) of four tokens (+2, +1, -1, or -4). (B–D). Shaded region indicates $\pm$ SEM, where N = the number of animals in each group (three VS lesion, four amygdala lesion, four control). RL = reinforcement learning; SEM = Standard error of the mean.

Figure 2

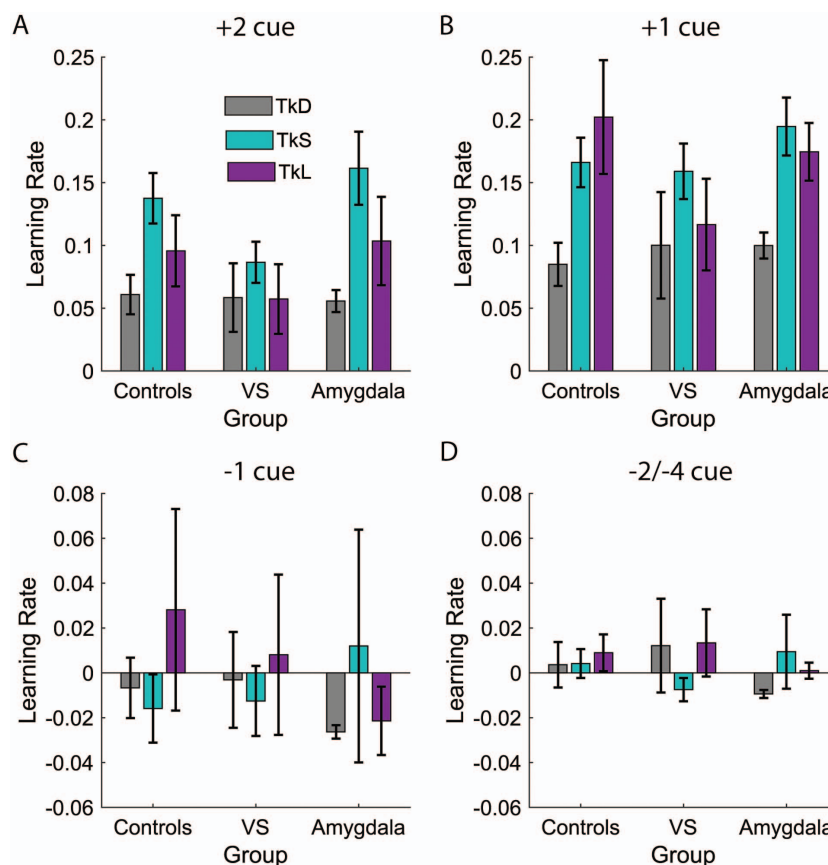
Within-Group Behavioral Performance Across the Three Tasks



Note. (A) Average performance for the control animals across tasks. (B) Average performance for the monkeys with VS lesions across tasks. (C) Average performance for the monkeys with amygdala lesions across tasks. Plots show the fraction of times monkeys chose the higher value option averaged across novel blocks for each group. Numbers at the top of each plot indicate the condition, which corresponds to the cues which were shown in those trials. (D) Average correct trials. Average correct images chosen by each group across tasks. Error bars are $\pm$ SEM, where N = the number of animals in each group (three VS lesion, four amygdala lesion, four control). VS = ventral striatum; TkD = deterministic tokens task; TkS = stochastic version of the tokens task; TkL = fourth tokens task; SEM = Standard error of the mean.

a cue that had occurred in the same location recently and the decay parameter measured the decay rate of the autocorrelation value. We only found group effects for the learning rate parameters, so for simplicity, we only discuss statistics for the learning rates.

When we compared learning rate parameters across all three experiments (Figure 3), we found that learning rates varied by cues, cue; $F(3, 24) = 83.1, p < .001$; experiments, experiment; $F(2, 14) = 5.8, p = .0150$; and for cues by experiments, Experiment $\times$ Cue: $F(6,$

Figure 3*Learning Rate Parameters for Each Cue Across the Three Tasks*

Note. (A) Learning rate parameters for the +2 cue across the three tasks. (B) Learning rate parameters for the +1 cue across the three tasks. (C) Learning rate parameters for the -1 cue across the three tasks. (D) Learning rate parameters for the -2/-4 cue across the three tasks. Error bars are $\pm$ s.e.m., where the N = the number of animals in each group (three VS lesion, four amygdala lesion, four control). VS = ventral striatum; TkD = deterministic tokens task; TkS = stochastic version of the tokens task; TkL = fourth tokens task.

43) = 4.7, p = .001. We also found that learning rates differed by group, group: $F(2, 8) = 5.2$, p = .037, and the difference in cues differed by group, Group $\times$ Cue: $F(6, 24) = 2.6$, p = .045.

To better characterize the group effects, we completed pairwise analysis of the groups. When we examined control and VS monkeys, we found that learning rates for cues differed by experiment, Experiment $\times$ Cue: $F(6, 27) = 5.1$, p = .001, but not across experiments, experiment: $F(2, 9) = 2.3$, p = .158. We also found a trend for control monkeys to have higher learning rates than VS monkeys across the experiments, group: $F(1, 5) = 7.9$, p = .039, but this effect did not survive correction. When we compared control and amygdala monkeys, we found that learning rates for cues differed across experiments and for different cues across experiments, experiment: $F(2, 10) = 5.4$, p = .025; Experiment $\times$ Cue: $F(6, 31) = 3.4$, p = .011. We found no group differences between control and amygdala monkeys, group: $F(1, 6) = 0.3$, p = .631. In the final comparison of amygdala and VS monkeys, we found that learning rates for cues differed across experiments, experiment: $F(2, 9) = 5.5$, p = .027, but not by experiment, Experiment $\times$ Cue: $F(6, 27) = 2.3$, p = .064. We found

a trend for amygdala monkeys to have higher cue learning rates than VS monkeys across the experiments, group: $F(1, 5) = 6.7$, p = .05, but this effect did not survive correction. We did, however, find that learning rates for cues differed by group, Group $\times$ Cue: $F(3, 15) = 4.7$, p = .016. Thus, there was a group effect on learning rates across all three groups. In other words, from the pairwise analyses between groups above, we found no effects between control and amygdala monkeys and marginal group effects between VS animals with either control or amygdala animals that did not survive correction. Therefore, there were differences among the three groups, and these could be marginally accounted for by lower learning rates in VS animals.

Next, we examined learning rates for the gain and loss cues separately (Figure 3). For the gain cues, we found differences between cues, cue: $F(1, 8) = 45.1$, p < .001, and experiments, experiment: $F(2, 14) = 17.6$, p < .001. We also found that the groups differed across gain cues, group: $F(2, 8) = 11.1$, p = .005. Learning rates for loss cues did not differ by experiment or group.

To investigate the group effect we found across gain cues, we completed pairwise analysis of the groups for learning rates for gain

cues across the three experiments. When we compared control and VS monkeys, we found that learning rates differed across cues, cue: $F(1, 5) = 32.2, p = .002$, and experiments, experiment: $F(2, 9) = 7.0, p = .014$. We also found a marginal effect for control monkeys to have higher learning rates when compared to VS monkeys, group: $F(1, 5) = 12.0, p = .018$. When we examined control and amygdala monkeys, we found that learning differed by cue, cue: $F(1, 7) = 24.7, p = .002$, and experiment, experiment: $F(2, 10) = 13.6, p = .001$, but not group, group: $F(1, 6) = 0.1, p = .727$. In the final group comparison, we compared learning rates for amygdala and VS monkeys. We found that learning rates differed by cue, cue: $F(1, 5) = 39.2, p = .002$, and experiment, experiment: $F(2, 9) = 22.3, p < .001$. We also found that amygdala monkeys had higher learning rates for gain cues when compared to VS monkeys, Group: $F(1, 5) = 49.7, p < .001$.

Motivation

Adaptive behavior requires that animals adjust their level of behavioral effort for their current environment. In essence, more effort requires more energy which is taxing leaving an organism tired. As a result, when it comes to operant behavior, animals must weigh the cost and benefits of being more or less effortful. In general, our behavioral analysis above already provides support for the hypothesis that this is what animals doing. When we looked at performance for all monks across the three tasks, we found that monkeys performed differently across the tasks, we also found that this level of performance across tasks varied by group. Our within-group analysis across the three tasks showed that this group difference was due to VS animals. VS animals were the only group to not perform statically different across the three tasks. This result suggest that VS animals were overall less sensitive to changes in the task environment. It is important to note that as far as information load goes, these tasks are identical. All three tasks have the same number of cues (4), conditions (6), and trials per block (108). Furthermore, animals saw both cues and conditions the same number of times in each block across the three tasks. This suggest that performance differences across tasks is not due to some information capacity issue (not due to ability), but instead due to the costs and benefits of giving more or less effort (motivation). These tasks only differed in three ways: reward schedule, reward rate, and absolute value difference of conditions. We hypothesized that performance across experiments varied with reward rate and the value difference in conditions across tasks. We can further solidify this cost–benefit theory of effort with two additional analyses based on how these two factors.

First, we looked at how the inclusion of the -4 cue in lieu of the -2 cue affected performance (condition value difference). While it is true that when we looked at average performance across TkS and TkL, we found no differences across tasks (Figure 2D). However, despite the fact that average correct trials were similar across these two tasks, performance across conditions indicated that this was likely due to the fact that three of the conditions have a larger value difference, due to the inclusion of the -4 cue (Figure 2A–C). To account for this, we looked at the two sets of conditions separately across the three tasks (Figure 4). The first set of conditions were the conditions that were the same across all three tasks (2 vs. 1, 1 vs. -1 , and 2 vs. -1). When we looked at this set of conditions, we found that monkeys performed differently across experiments, experiment: $F(2, 14) = 45.3, p < .001$ and that this difference varied across groups, Group $\times$ Experiment;

$F(4, 14) = 3.9, p = .022$. Pairwise analysis revealed that this effect was driven by VS animals not modulating their behavior like the other two groups across this set of conditions, controls versus VS: Group $\times$ Experiment; $F(2, 9) = 7.9, p = .01$; amygdala versus VS: Group $\times$ Experiment; $F(2, 9) = 4.9, p = .03$; controls versus amygdala: Group $\times$ Experiment; $F(2, 10) = 0.5, p = .62$.

When we looked at the second set of conditions, the ones with the larger value difference (1 vs. $-2/-4$, 2 vs. $-2/-4$, -1 vs. $-2/-4$), we found that monkeys performed differently across experiments, Experiment; $F(2, 14) = 64.4, p \leq .001$. There were no group effects across this set of conditions. This analysis revealed that all groups modulated their behavior in a similar fashion across the conditions that had a value difference. This result taken with the comparison of conditions that were the same across all tasks suggests that groups are similarly sensitive to changes in cue value, and the difference between groups across tasks is not due to the change in the absolute value difference of conditions.

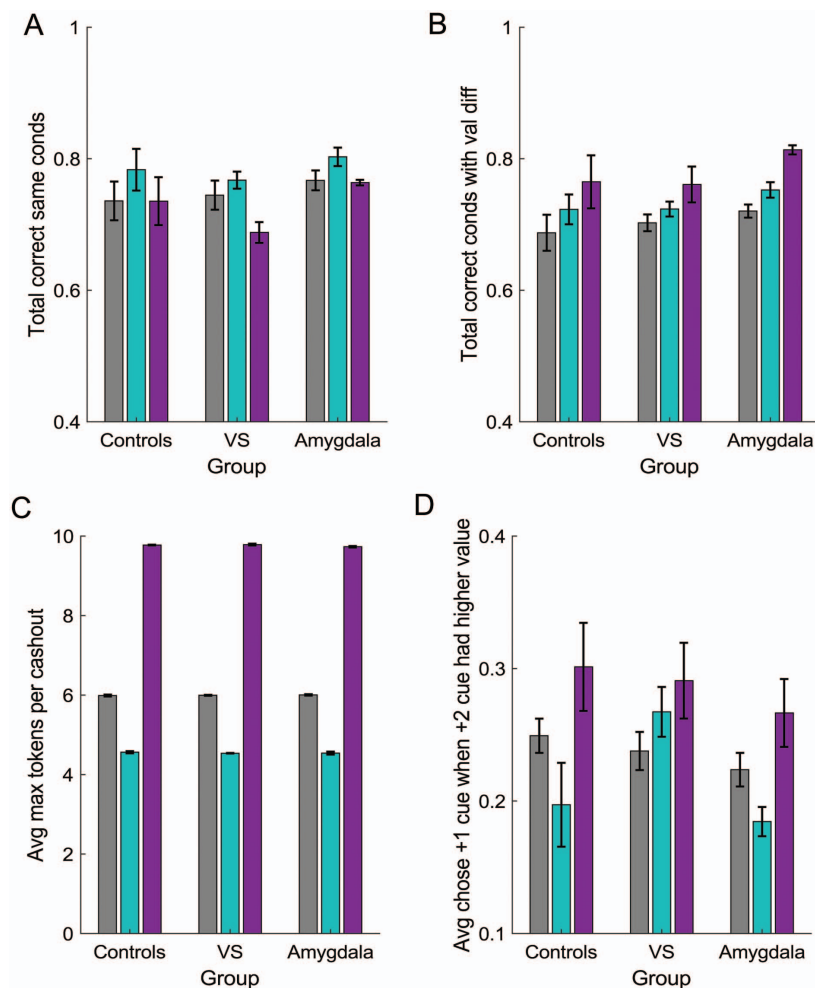
The distinction between motivation and learning is about the distinction between what one can do (ability) VS what one will do (motivation). As we stated earlier, there are only four cues per block, thus the $+2$ cue in the 2 versus 1 condition is the same as the $+2$ cue in the 2 versus -1 and 2 versus -2 conditions. Monkeys always perform better in the latter two conditions. This is interesting because conditions were presented in pseudorandom order. Monkeys saw each condition twice, once on the left and once on the right, every 12 trials before seeing any condition a third time. This pseudorandom order was followed throughout the block, which makes it interesting that performance in the 2 versus 1 condition lags behind the other two conditions that have the $+2$ cue in them. If monkeys know the $+2$ cue in the 2 versus -1 and 2 versus $-2/-4$ conditions by Trial 6, there is no reason other than motivation for them performing worse in the 2 versus 1 condition by the eighth time they see this condition (at the latest).

We hypothesized that when everything else is held constant, it was the task reward rate that affected motivation and led to performance differences across tasks. These three tasks had different reward rates, with TkL having the richest reward environment, and TkS having the leanest reward environment (Figure 4). The fact that performance in the 2 versus 1 condition seemed to vary with the reward rate of the task, with performance being the best in TkS, and performance being the worse in TkL is additional support that we are looking at motivation. We can provide further support for this notion by analyzing the choice probabilities from the RL model. In essence, choice probabilities show what information animals had at the time of their choice. When we looked at the average proportion of time animals selected the $+1$ cue in the 2 versus 1 condition, when the $+2$ had a higher value (Figure 4), we found that this proportion differed across tasks, experiment: $F(2, 14) = 13.7, p < .001$. We also found that groups differed across tasks, Group $\times$ Experiment; $F(4, 14) = 5.2, p = .008$. Pairwise analysis revealed that this effect was driven by VS animals not modulating their behavior like the other two groups across this set of conditions, controls versus VS: Group $\times$ Experiment; $F(2, 9) = 7.4, p = .012$; amygdala versus VS: Group $\times$ Experiment; $F(2, 9) = 12.2, p = .002$; controls versus VS amygdala: Group $\times$ Experiment; $F(2, 10) = 0.7, p = .521$.

Discussion

In this study, we examined factors that drove task-specific deficits in choice behavior in monkeys with VS lesions. We found that the

Figure 4
Motivation Across the Three Tasks



Note. (A) Average correct trials for the conditions that were the same across the three tasks. (B) Average correct trials for the conditions that differed across the three tasks, due to the larger loss (−4) cue in TkL. (C) Maximum number of tokens monkeys could earn on average per cash-out across the three tasks. (D) Average proportion of times monkeys chose the +1 cue when the +2 cue had a higher choice probability (value) across the three tasks (these values came from the RL model). Error bars are $\pm$ SEM, where N = the number of animals in each group (three VS lesion, four amygdala lesion, four control). TkL = fourth tokens task; RL = reinforcement learning; VS = ventral striatum; SEM = Standard error of the mean.

deficits for VS animals in TkS and TkL are due to VS monkeys not modulating their choice behavior as well as control and amygdala monkeys in these tasks. Specifically, in TkS, VS animals did not significantly increase their performance, while the other two groups did. This extends our previous work examining the role of VS in learning (Taswell et al., 2018), where we found that monkeys with VS lesions had performance deficits in TkS and TkL, but not TkD. Here, we also report results in which we found that monkeys with amygdala lesions had no performance differences relative to controls in any of the experiments.

When we compared performance in TkD to TkS, we found that all monkeys performed better in TkS. Our within-group analyses revealed that both the control and amygdala lesion monkeys performed

significantly better in TkS, while the VS monkeys only trended toward better performance in TkS. This point, taken with the fact that we found no differences between control and VS monkeys in TkD, revealed that the behavioral deficits in the VS animals in TkS (Taswell et al., 2018) were due to control monkeys improving their choice behavior in this task, rather than VS monkeys worsening their choice behavior. This analysis also explains why we found no performance differences between amygdala lesion and control animals in TkD or TkS, which follows because the amygdala monkeys also improve their choice behavior in TkS.

When we compared TkD and TkL, we found that monkeys performed better in the TkL experiment. Our within-group analysis revealed that control and amygdala monkeys performed better in

TkL. However, this was not the case for VS monkeys. Thus, the difference between the groups in this comparison is that monkeys with VS lesions chose less accurately in TkL. This can be seen in the three conditions without a value difference. While all groups chose less accurately in the 2 versus 1 condition, VS monkeys also chose less accurately in the 1 versus -1 and 2 versus -1 conditions (Figure 2B).

We also used RL models to characterize the factors driving the differential performance across experiments. We found that VS animals had lower learning rates for gain cues in TkS and TkL, when compared to control and amygdala monkeys. Specifically, in TkS, VS monkeys do not learn values as quickly for the +2 cue relative to the other groups. While in TkL, VS monkeys do not learn values for either gain cue (+2, +1), as fast as the other two groups. These cue-specific learning rates may reflect the values of the cues in their environmental setting. This is illustrated by the fact that monkeys performed better in TkS, when compared to TkD. TkD and TkS only differed in two ways. One, TkS employed a stochastic reward environment, while TkD had a deterministic reward environment. Two, because TkS used a stochastic reward environment, the reward rate in TkS was lower than the reward rate in TkD. In general, if this effect is due to learning, and not motivation, it should be more difficult to learn a stochastic cue-reward association when compared to a deterministic cue-reward association. Therefore, the better performance we found in TkS, and thus, the increase in learning rates for the +2 cue in the TkS (when compared to TkD) may reflect motivational value. Notably, this motivational value might manifest as attention. Since paying attention is effortful, more motivated animals may pay closer attention to their environment. Specially, it would require more motivation and attention to perform better in stochastic environments. Additionally, because monkeys with amygdala lesions performed as well as controls, the effect of the VS lesions on behavior is unlikely to be due to nonspecific effects of surgery (e.g., anesthesia, two-stage operation).

Our condition and choice probability analysis further support this notion of motivational value and revealed that reward rates, which reflect reward schedules, affect performance. Importantly, this effect showed that when other task factors were held constant, except for reward structure and reward rate, animals tended to improve their performance in the task with the leaner reward environments (TkS). As we stated earlier, it should be more difficult to learn a stochastic cue-reward association (TkS) when compared to a deterministic cue-reward association (TkD). Yet, our analysis shows that despite animals having less feedback in TkS compared to TkD, animals performed better in TkS. We submit that because rewards were rarer in TkS, animals more often selected the better option in the more difficult learning environment (TkS). Recent work on dopamine neurons provides support for this result. Rothenhoefer et al. (2021) found increased dopamine responses in monkeys receiving rare rewards when compared to monkeys receiving constant rewards. Importantly, this increased dopamine response was not due to prediction errors as they found this amplified response for rare rewards even when prediction errors were equal.

We have shown that monkeys complete the tokens task at varying levels of performance. We hypothesize that this difference in performance is due to environmental factors that shape motivation. Our results along with findings in the literature support the idea that the VS and amygdala are both critical for motivation. Notably, our results suggest that the VS and amygdala are important for distinct

types of motivation. More specifically, we hypothesize that the VS is important for motivation that affects the amount of effort animals will give in different types of learning environments. Through results presented here, we show that monkeys with VS lesions had lower ceiling (TkS) and lower floor (TkL) performance, when compared to the other two groups. This result suggests a role for the VS in motivational effort because monkeys with VS lesions modulated their behavior in the same direction as the other two groups across tasks but did not modulate their performance as well as the other two groups. This theory is supported by recent work that found neural activity in the VS representing different learning environments or contexts (Gmaz & van der Meer, 2022). Importantly, this form of context signal was found to be distinct from reward-related value. Furthermore, this context signal was found to predict modulations in behavioral responses to reward-predictive cues. This work demonstrates a mechanism for encoding context-dependent modification of behavior. Additionally, all mice displayed a ramping signal conveying expected reward value in each context. The VS could process both the context signal and the motivationally relevant value to signal different motivational contexts and to correspondingly modulate behavior. The type of motivation that the amygdala is important for is less clear. From our results and the literature, we suggest that the amygdala is important for motivation in concurrent appetitive learning environments, where two cues are rewarded with the same magnitude at different probabilities. We explore and provide support for these theories below.

Many of the learning studies that have found a role for dopamine, the VS, or the amygdala employed similar reward structures. These studies used concurrent appetitive reward structures, in which choice of either cue is rewarded, but at different rates (Costa et al., 2016; Eisenegger et al., 2014; Pessiglione et al., 2006). For example, reward rates of 80/20, 70/30, and 60/40 for the two cues have frequently been used. Current RL theories assume that this reward structure is a more difficult learning environment, compared to a pure stochastic learning environment that rewards choices using a 75/0 schedule, in which one cue is rewarded 75% of the time and the other cue is never rewarded. In essence, this view assumes that animals are rational agents trying to identify and select the best choice option, independent of the reward environment. Our results, however, suggest there is a differential effort cost between concurrent appetitive, pure stochastic, and deterministic learning environments.

Another perspective, distinct from rational optimization theory, is that animals find concurrent appetitive schedules less motivating than pure stochastic schedules. Since animals have a chance of receiving rewards with either option in the concurrent environment and not in the pure stochastic, or deterministic environment, we believe that animals should be more motivated in the pure stochastic, or deterministic environment. In essence, we are suggesting that concurrent appetitive learning environments drive down motivational value because rewards are easier to obtain (rich reward environment). In standard operant contingencies that have rich reward environments, it has been found that the optimal solution is to respond faster for all possible actions (Niv et al., 2007). That is to say that the optimal latencies of all actions are inversely proportional to the average reward rate. A rich reward environment is often defined as an environment where it does not matter what the animal does—rewards are plentiful and the effort cost is low. The reward environment used explains how the monkeys in this study sometimes show remarkably different levels of performance across tasks.

The task change is likely manipulating motivation rather than measuring ability.

To solidify this concept further, consider two additional studies that used the same VS (Rothenhoefer et al., 2017) and amygdala lesion monkeys (Taswell et al., 2021) in a two-armed bandit reversal learning task. In these studies, the VS and amygdala were found to be important for stimulus-based RL. However, as our current analyses demonstrate, these same animals can learn stimulus-based RL as well as controls. While the what/where task used previously (Rothenhoefer et al., 2017; Taswell et al., 2021) differs in several ways from the token tasks described in this article, we can say that if either of these structures was crucial for the acquisition of the stimulus–consequence relationship, we should consistently find deficits in these animals.

These results together suggest that the inconsistent performance found for animals across tasks may be due partially to motivation and not the effects of lesions on learning ability per se. We propose that this is the reason learning studies that have employed deterministic reward structures have found no performance deficits for VS (Costa et al., 2016; Vicario-Feliciano et al., 2017) or amygdala (Izquierdo & Murray, 2007; Málková et al., 1997) lesion animals. In studies that use deterministic learning environments, the cost of each missed reward is higher than in stochastic concurrent appetitive reward structures because the worse cue is never rewarded. This motivates animals to increase their effort, so that they can track and pick the rewarded cue. Our current results support and shed further light on this process. Our results showed that animals were sensitive to the reward structure and reward rate. When reward rates are high in deterministic reward structures, the amount of effort animals put forth in the task was also affected. This may be why we found that VS monkeys had performance deficits in TkL (Taswell et al., 2018). The endowment we gave monkeys seemed to make them value the gain cues less in certain conditions. This is particularly true for the monkeys with VS lesions, but it can be seen to some degree for all monkeys in the 2 versus 1 condition. In general, across all of the tasks, monkeys performed the worst in the 2 versus 1 condition in TkL. In contrast, monkeys performed the best in the 2 versus 1 condition in TkS, which had the leanest reward structure.

The literature on effort provides further support for the idea that animals are not simply trying to get the maximum number of rewards regardless of differences in effort. In a task where rats were given the choice between an easy or hard discrimination, where successful hard trials were reinforced with double the reward (Cocker et al., 2012), researchers found individual variations in rats' willingness to work. The term "workers," refers to rats that naturally chose the hard trials significantly more than other rats, called "slackers." In this study, researchers found when slackers were given amphetamines, they would choose the hard trials more often. Recent work in humans (Westbrook et al., 2020) is consistent with this finding. In this study, the authors found that "workers" naturally had higher striatal dopamine levels, when compared to "slackers." Amphetamines given to slackers produced similar levels of striatal dopamine and lead to "slackers" choosing the hard trials as often as "workers."

This individual variation in willingness to expend cognitive effort is important because it shows that if animals are given a choice, some will prefer a low effort–small reward option over a high effort–high reward option, which could be the case in the concurrent appetitive reward structures. Regardless of individual variation,

studies on effort have consistently found that dopamine depletion from the VS or pharmacological manipulations of dopamine in the VS leads to deficits on higher fixed ratio schedules (Aberman & Salamone, 1999; Ishiwari et al., 2004; Koch et al., 2000; Mingote et al., 2005). There are two important facts about these studies. First, there were no deficits in the dopamine-depleted rats on an FR1 schedule, but as the schedules increased (FR4, FR16, FR64), so did the deficits. Second, these studies showed that the dopamine depletion did not affect primary food reinforcement. For free or low effort cost, animals with VS dopamine depletion consumed as much as control counterparts. These results suggest that the VS is important for the amount of effort animals are willing to put forth. Similar results have been found in delayed discounting tasks, where waiting for the larger reward is thought to be more effortful. Animals given haloperidol or animals that received lesions of the VS choose small immediate reward more often than controls (Cardinal et al., 2001; Cardinal & Howes, 2005; Denk et al., 2005).

Adaptive behavior requires animals to not only be sensitive to the reward rate and reward structure but also to the effort required in these environments. The way in which all the monkeys in the present study alter their performance suggests that animals have an innate mechanism that takes environmental factors into account and decides how much effort should be devoted. Our results, along with the work on effort discussed above, provide support for the VS playing a motivational role in RL. While both the amygdala and VS have been implicated in operant responding, their contributions appear to be distinct and mirror our results. One study showed that when rats received lesions of the basolateral amygdala, which has been shown to lead to a reduction in instrumental responding (Everitt et al., 1989), and then received intra-accumbens amphetamine infusions, there was a dose-dependent increases in responding. Importantly, these intra-accumbens amphetamine infusions made the rats with basolateral amygdala lesions respond like controls (Cador et al., 1989).

This result should be viewed in contrast to another study (de Borchgrave et al., 2002) that compared rats with lesions of the VS to controls and found that instrumental responding was mildly suppressed in the rats with VS lesions. However, the authors point out that this reduction in responding was not due to lesion animals having trouble learning the instrumental contingency, but instead due to a reduction in motivation. In a second experiment, amphetamines were administered to control and lesioned rats, and the normally increased response found when amphetamines were administered was significantly reduced in lesion animals.

The fact that amphetamine administration made rats with basolateral amygdala lesions, but not rats with VS lesions, respond like controls provides further support for the finding that VS dopamine is important for making animals more sensitive to rewards and less sensitive to effort cost (Westbrook et al., 2019). We suggest that animals have an innate ability to shift their motivational state based on the reward structure and effort cost of the learning environment. Dopamine and the VS are critical to this shift in motivation (Hamid et al., 2016; Niv et al., 2007). This view explains the effort literature stated above and why we find deficits in VS animals in some of the token tasks, but find no deficits in amygdala animals. While we cannot say for sure which task factor made monkeys with amygdala lesions modulate their effort like controls, theoretically, it could be the addition of loss or the addition of conditioned reinforcers (tokens) to the learning environment. Future studies could address

these possibilities. Regardless, it appears that an intact VS is important for this level of effort modulation. The monkeys with amygdala lesions still have their VS, which along with dopamine allows them to modulate their choice behavior to the level of controls across the token tasks.

Conclusion

To differentiate aspects of learning from environmental factors that affect motivation, we ran a series of experiments with varying task factors. We investigated the involvement of the VS and amygdala across these experiments. We were able to show that task environment is a significant factor that influences choice behavior. Notably, our finding does not fit current RL theories. We found that reward rate and reward structure can affect the level of motivation animals have in a particular learning environment, and this level of motivation affects how much effort animals are willing to put forth in said learning environment. Our results indicate that the VS is important for effort modulation across learning environments. Our within-group analysis across tasks revealed four important findings. One, all groups modulated their behavior in the same direction across the tasks, showing how learning environments can elicit different levels of motivation, and this level of modulation is the key to understanding our findings. Two, the reason we find deficits in VS animals in TkS was not due to VS animals having difficulties learning in stochastic learning environments but instead due to control and amygdala animals performing better in the stochastic learning environment. Three, the reason we find deficits in VS animals in TkL when compared to TkD, the task, where VS animals had no performance deficits, is due to VS animals performing worse in the first three conditions in the rich deterministic learning environment. Four, we show that animals with amygdala lesions can learn stimulus-based RL as well as controls in learning environments with loss and conditioned reinforcers. These findings likely account for some of the conflicting reports regarding contributions of the VS and amygdala to learning. In addition, our findings connect the literature on motivation with RL to better explain how animals adapt in environments with different reward rates and structures that require varying effort.

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